

LLM Extraction of Interpretable Covariates

Structured EHR (observed)

- Demographics
- Vital signs
- Labs
- Diagnoses (ICD)
- Procedures
- Medications

Clinical Notes (unstructured)

"...Frailty and poor functional status.
DNR/DNI, comfort measures only.
Waxing and waning mental status..."

Structured Covariates



Functional status
Mental status
Code status
Infection source
Substance use
+ more ...

Covariate Integration Strategies

Baselines (Text Representations)

- M1** Tabular Only
(Structured EHR only)
- M2** TF-IDF + PS
(Traditional NLP)
- M3** BioClinicalBERT + PS
(Embedding)

Propensity Score Modeling



(Logistic Regression)

LLM-Augmented Strategies

- M4** Direct PS Augmentation
 $PS(T | X_{tab}, X_{LLM})$
- M5** Dual-Caliper Matching
(PS caliper + Text distance)
- M6** IPW Weighting
(Using IPW with X_{tab}, X_{LLM})

- M7** Full PSM
(Sensitivity) all 7 LLM covariates

Validation Framework

Balance Assessment

(Standardized Mean Difference)



Lower SMD
is better

Semi-Synthetic Experiments

(Known Ground-Truth ATE)



Evaluate Bias
& RMSE

Real-World Application

ATE & HTE Subgroup Discovery



Robustness &
Interpretability

Key Findings

1 Best Covariate Balance

Mean SMD across 26 covariates (lower is better)

M1 Tabular Only

M2 TF-IDF + PS

M3 BioClinicalBERT + PS

M5 Dual-Caliper Match

M6 IPW Weighting

M4 Direct PS Augmentation

Best
Balance!

2 Lowest Bias in Semi-Synthetic Data

Bias of ATE estimate (vs. true $\tau = -0.05$, lower is better)

M1 Tabular Only

0.014

M2 TF-IDF + PS

0.011

M3 BioClinicalBERT + PS

0.008

M5 Dual-Caliper Match

0.014

M6 IPW Weighting

lower bias
vs. Tabular Only!

0.0012

M7 Full PSM

0.0005

M4 Direct PS Augmentation

0.0003

3 Discovers New Effect Heterogeneity

Top Effect Modifiers Identified Using LLM Covariates

Covariate

Strength of Heterogeneity (Treatment effect difference)

Code status (DNR/DNI)

★★★★★

Infection source

★★★★☆

Functional status

★★★★☆

Mental status

★★★★☆

.....

New clinical
subgroups
uncovered!